

Keywords: recurrent neural networks • depth vs. width • compute allocation • sequence modeling • neural scaling

Temporal Recurrence Favors Fewer Layers

Recurrent Models Peak at 2–4 Layers While Non-Recurrent Models Keep Improving Up to 32 — Temporal Depth and Within-Step Depth Provide Overlapping Computational Benefits

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arXiv:2609.12531

Deep learning lore holds that depth is essential for model expressivity. This intuition was developed for non-recurrent models that process all sequence positions in parallel and must instantiate all computation simultaneously. Recurrent architectures add a qualitatively different computation axis: temporal recurrence, which accumulates processing across sequence positions step by step. This paper asks: when temporal recurrence is available, how much within-step depth is still needed? The answer is surprisingly little — and the difference has direct implications for how recurrent models should be scaled.

AT A GLANCE

Problem: How should compute be allocated across within-step depth, expert width, and parallel experts for recurrent vs. non-recurrent models?

Why it matters: Miscalibrated depth wastes compute that could go to width; the right allocation changes at optimal recurrent model design.

Finding: Recurrent models peak at 2–4 layers; non-recurrent models keep improving through 8–32 layers.

Explanation: Temporal recurrence and within-step depth both allow later computation to condition on earlier outputs — recurrence makes depth partly redundant.

Evidence: Sokoban planning and FineWeb language modeling; matched FLOPs across depth, width, expert-count axes.

The Big Picture

A non-recurrent Transformer of depth D' processes each token through D' layers applied once in parallel. A recurrent model of within-step depth D applied over T sequence positions uses $D \cdot T$ total layer applications. Under matched FLOPs, the question is: for a fixed $D \cdot T$ budget, should D be large (deep per-step, fewer steps of computation) or small (shallow per-step, many recurrent

steps)?

The answer the authors find empirically: small D with compensating width is consistently better for recurrent models, while large D is better for non-recurrent models. This suggests that within-step depth and temporal depth are not fully orthogonal dimensions — they provide partly overlapping computational benefits.

Technical Formulation

Let $\mathbf{h}_t^{(0)} = \mathbf{x}_t$ be the input at position t . The recurrent forward pass applies D within-step layers:

$$\mathbf{h}_t^{(d)} = f_d(\mathbf{h}_t^{(d-1)}, \mathbf{h}_{t-1}^{(D)}), \quad d = 1, \dots, D$$

where $\mathbf{h}_{t-1}^{(D)}$ is the hidden state from the previous position, and weights are shared across positions (weight tying across time). For E parallel experts:

$$\mathbf{h}_t^{(d)} = \sum_{e=1}^E \pi_e(\mathbf{h}_t^{(d-1)}) \cdot f_d^{(e)}(\mathbf{h}_t^{(d-1)})$$

Compute matching: When depth is reduced from D to D/k , the hidden dimension is increased by $\sqrt{k}$ to hold parameters constant (quadratic scaling of attention/MLP with hidden dim).

Arithmetic intensity: Total layer applications per sequence equal $D \cdot T$ (recurrent) vs. D' (non-recurrent, applied once to all positions in parallel). FLOPs are matched across configurations.

What the Results Show

Sokoban Planning (Success Rate, %)

Depth D	Recurrent	Non-Recurrent
1	44.2	31.8
2	67.9	49.3
4	71.3	58.7
8	70.1	66.4
16	68.4	70.8
32	66.7	72.1

Recurrent models peak at $D = 4$; non-recurrent models keep improving through $D = 32$. At the recurrent optimum ($D = 4$), the recurrent model outperforms the non-recurrent model by 12.6 points at the same FLOPs.

FineWeb Language Modeling (Validation Perplexity, ↓)

Depth D	Recurrent	Non-Recurrent
2	18.7	22.4
4	17.3	20.1
8	17.8	18.6
16	18.4	17.9
32	19.1	17.2

Consistent finding: recurrent optimal at $D = 4$; non-recurrent at $D = 16$ –32.

Ablations and Interpretation

Width compensation: Reducing recurrent depth from 8 to 2 layers while increasing hidden dimension to match FLOPs recovers 97% of 8-layer recurrent performance on both tasks. Width is a suitable substitute for depth in recurrent networks.

Expert count: Increasing parallel experts from 1 to 4 (same compute via width reduction) provides consistent modest gains for both architectures, orthogonal to the depth effect.

Shared vs. unshared weights: Unshared weights slightly improve non-recurrent baselines but do not change the qualitative finding for recurrent models.

Sequence length effect: Longer sequences amplify the advantage of shallow recurrent models over deep ones, consistent with the interpretation that temporal depth substitutes for within-step depth; more recurrent steps make additional layers more redundant.

Reference

Ivan Anokhin, Johan Obando-Ceron, Irina Rish, Sebastian Risi. **Temporal Recurrence Favors Fewer Layers.** arXiv:2609.12531, September 2026. <https://arxiv.org/abs/2609.12531>